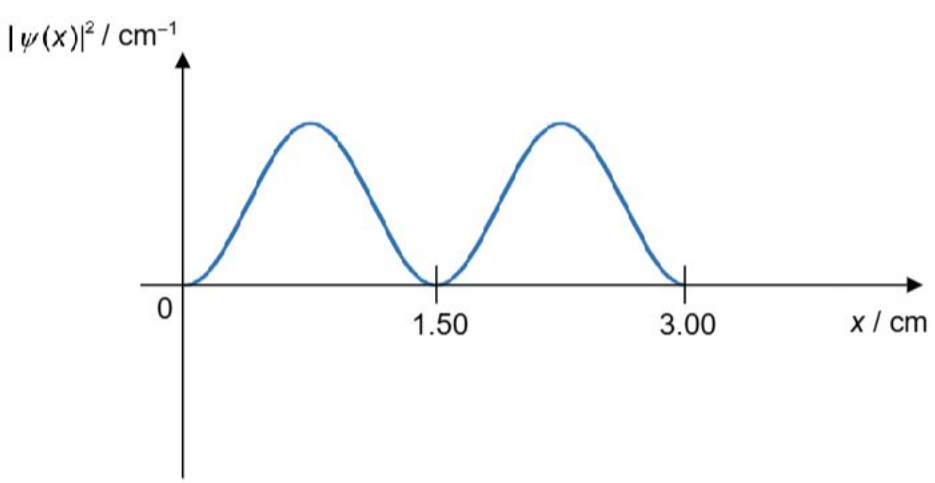
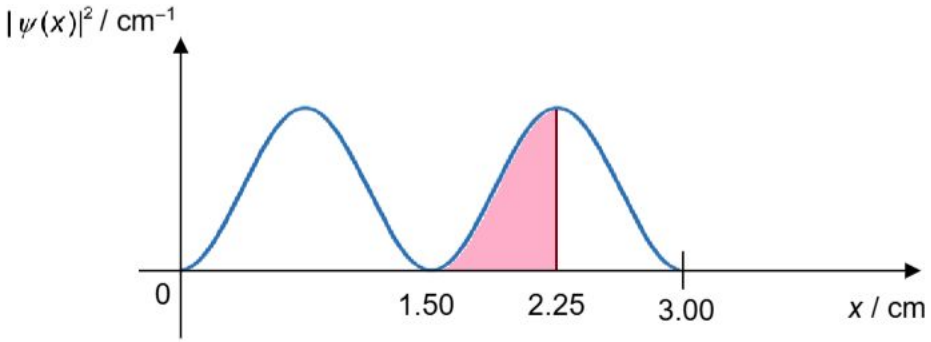
7(a)(i)	In the photoelectric experiment, electrons are observed to only be emitted when the surface is irradiated with <u>electromagnetic radiation</u> of frequency <u>above the threshold frequency</u>. This suggests that the <u>interaction</u> between an electron and a photon is <u>one-to-one</u>. The <u>absorption of **a** photon leads to a gain in energy for **an** electron</u> to be emitted.	B1 B1
7(a)(ii)	The formation of the interference pattern on the screen shows that electrons can <u>diffract through a slit</u>. Electrons can also undergo <u>constructive and destructive interference</u> to form regions of maxima and minima.	B1 B1
7(b)(i)	Allowed energy E for a particle in a box is given by $$E_n = \frac{h^2}{8mL^2}n^2$$ 'first excited state' → $n = 2$ 'ground state' → $n = 1$ Loss in energy of electron during transition from n=2 to n=1 $\Delta E_n = \frac{h^2}{8mL^2}(n_i^2 - n_f^2) = \frac{(6.63\times10^{-34})^2}{8(9.11\times10^{-31})(0.0300)^2}(2^2 - 1^2)$ $= 2.01\times10^{-34}$ J Comments: A large number of students incorrectly stated that the energy of the electron at ground state is 0 J.	M1 A1
7(b)(ii)	$E = hf = \frac{hc}{\lambda}$ $2.01\times10^{-34} = \frac{(6.63\times10^{-34})(3.00\times10^{8})}{\lambda}$ $\therefore \lambda = 9.90\times10^{8}$ m	
7(b)(iii) 1		A1
7(b)(iii) 2	area under graph = 1	M1

	$\left(\frac{A^2}{2}\right)(3.00) = 1$ $A = 0.816 \text{ cm}^{-\frac{1}{2}}$	A1
7(b)(iii) 3	 ∴ probability = 0.250	A1
8(a)(i)	mass defect $\Delta m = 2 \times 1.007825u + 2 \times 1.008665u - 4.002604u =$	M1